

Exercise-1: Control Structures

exercise1.sql:

-- Step 1: Create Tables

```
CREATE TABLE Customers (  
    CustomerID NUMBER PRIMARY KEY,  
    Name VARCHAR2(100),  
    Age NUMBER,  
    Balance NUMBER,  
    IsVIP VARCHAR2(5)  
);
```

```
CREATE TABLE Loans (  
    LoanID NUMBER PRIMARY KEY,  
    CustomerID NUMBER,  
    InterestRate NUMBER,  
    DueDate DATE,  
    FOREIGN KEY (CustomerID)  
        REFERENCES Customers(CustomerID)  
);
```

-- Step 2: Insert Data

```
INSERT INTO Customers VALUES (1,'Aditya',65,15000,'FALSE');  
INSERT INTO Customers VALUES (2,'Rahul',45,8000,'FALSE');  
INSERT INTO Customers VALUES (3,'Priya',70,20000,'FALSE');
```

```
INSERT INTO Loans VALUES (101,1,8.5,SYSDATE+20);  
INSERT INTO Loans VALUES (102,2,9.0,SYSDATE+50);  
INSERT INTO Loans VALUES (103,3,7.5,SYSDATE+10);
```

```
COMMIT;
```

```
-- Step 3: Scenario 1
```

```
BEGIN
```

```
  FOR cust IN (
```

```
    SELECT CustomerID
```

```
    FROM Customers
```

```
    WHERE Age > 60
```

```
  )
```

```
  LOOP
```

```
    UPDATE Loans
```

```
    SET InterestRate = InterestRate - 1
```

```
    WHERE CustomerID = cust.CustomerID;
```

```
  END LOOP;
```

```
  COMMIT;
```

```
END;
```

```
/
```

```
-- Step 4: Scenario 2
```

```
BEGIN
```

```
  FOR cust IN (
```

```
    SELECT CustomerID
```

```

        FROM Customers
        WHERE Balance > 10000
    )
    LOOP
        UPDATE Customers
        SET IsVIP = 'TRUE'
        WHERE CustomerID = cust.CustomerID;
    END LOOP;
    COMMIT;
END;
/

-- Step 5: Scenario 3
SET SERVEROUTPUT ON;

BEGIN
    FOR loan_rec IN (
        SELECT c.Name, l.LoanID, l.DueDate
        FROM Customers c
        JOIN Loans l
        ON c.CustomerID = l.CustomerID
        WHERE l.DueDate BETWEEN SYSDATE AND SYSDATE + 30
    )
    LOOP
        DBMS_OUTPUT.PUT_LINE(
            'Reminder: Dear ' ||

```

```
        loan_rec.Name ||  
        ', Loan ' ||  
        loan_rec.LoanID ||  
        ' is due on ' ||  
        TO_CHAR(loan_rec.DueDate,'DD-MON-YYYY')  
    );  
END LOOP;  
END;  
/
```

-- Step 6: Verify Results

```
SELECT * FROM Customers;
```

```
SELECT * FROM Loans;
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Table CUSTOMERS created.

Table LOANS created.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

Commit complete.

PL/SQL procedure successfully completed.
```

PROBLEMS	OUTPUT	DEBUG CONSOLE	TERMINAL	PORTS	GITLENS	SONARQUBE
All rows fetched: 3 in 0.023 seconds						
	CUSTOMERID	NAME	AGE	BALANCE	ISVIP	
1	1	Aditya	65	15000	TRUE	
2	2	Rahul	45	8000	FALSE	
3	3	Priya	70	20000	TRUE	